CIS 2267 Midterm Project

Learning Objectives

- Finding Data Needed.
- Determining the Accuracy of the Data and its Origins.
- Legal/Ethical Issues in Using the Data.
- Doing Due Diligence on the Data.
- Using Computational Techniques to Analyze and Visualize Data.

My Data Quest

Situation: You are a student at Carlini's College. You are writing a term paper in your Sustainability course. You have chosen to study the decrease in USA farm acreage resulting from ongoing urbanization. You are especially interested in Ohio as you reside there, it is an agricultural state, with seven major metropolitan areas expanding rapidly, especially Columbus. This topic touches many contemporary diversity, equity, and inclusion issues, including:

- United Nations Sustainable Development Goals (#2 Zero Hunger)
- Food insecurity
- Malnutrition
- Starvation
- Equity
- Inclusion
- Social justice

together with international, national, state, and local governmental policy implications.

You are doing Internet research to locate useful information and data to support your term paper's thesis and recommendations. You are concerned about finding data that:

- Either supports or refutes your arguments
- The source and accuracy and unbiasedness (the data doesn't have an agenda) of the data
- Its understandability to readers
- Whether it lends itself to the types of analysis you want to make
- Finally, you are concerned about your ability to ethically and legally use the data you find

Below represents the author's journey from identifying a data need to finding the data. It is intended to show the author's thought process in hopes that it will help you improve your Methods of Collecting Data.

My mission is to locate one specific dataset:

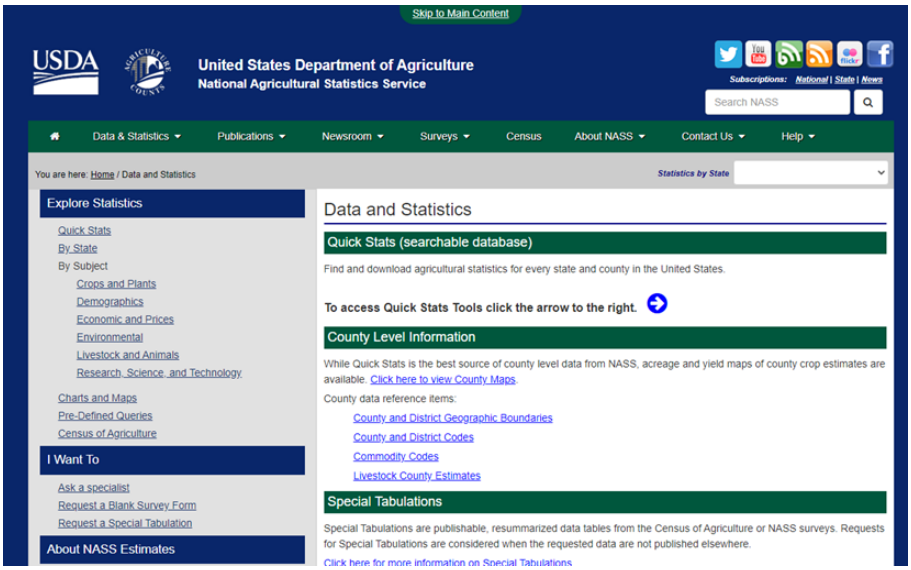
Most recent year's farm acreage compared to a previous year, by US state.

Note: This is a long document! There is not much for you to do initially, but the process of finding and judging data is an important part of data analytics. Data analytics is more than just manipulating data and generating visualizations. Data analytics requires skills in internet searching, evaluating data sources, and critical thinking. Read the author's journey, understand the thought process, and apply what you learn to your own data-gathering processes.

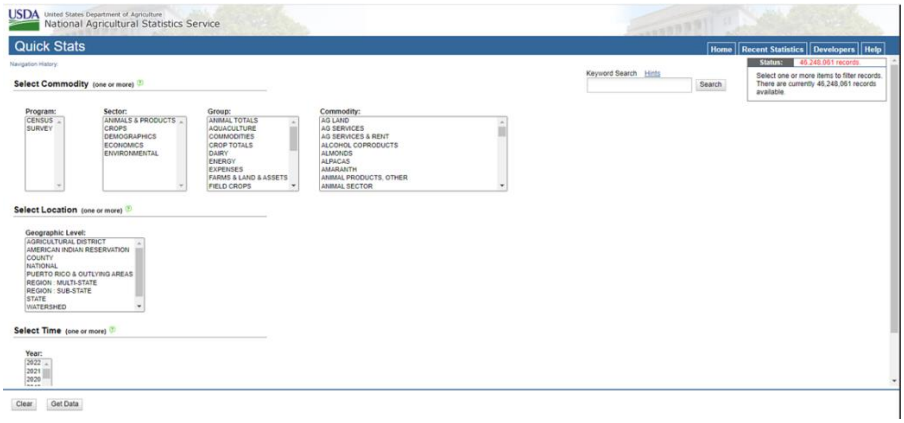
Where to Start?

Knowing that US government data is usually freely usable in any legal/ethical way, I decided to start with the US Department of Agriculture (USDA) website. I performed an internet search for "USDA Data". Among the search results, I found:

https://www.nass.usda.gov/Data_and_Statistics/



And then I clicked on Quick Stats (blue right-arrow in screenshot above). See the screenshot below for results. I guess the "Quick" in Quick Stats is that I was *quickly* overwhelmed by 42 million records (rows) and hundreds of drop-down menu choices. Mine eyes glaze over. After experimenting a bit, I *quickly* realized that I could not *quickly* get up to speed and find the data I wanted. I became *quickly* frustrated. So, I moved on to more searching. (Please note, this is not an endorsement for simply giving up because it's challenging.)



NASS = National Agricultural Statistical Service. Keep that in mind. We'll encounter it later.

I also found out that USDA agricultural census data is developed every five years, in years ending in "2" and "7". That fact, too, will be useful later.

Next Steps

My next thought was that maybe individual state agricultural-data websites might be less intimidating. So, I tried Ohio and Indiana (you can search for them yourself if you're interested). After exploring, I realized that the origin of their data was essentially NASS data repackaged a bit. Applying reasoning, I figured no state is going to independently create some subset of 42 million records when they can simply access their own state's NASS data. Furthermore, the thought of finding farmland acreage data on fifty separate state websites was not appealing.

I continued my searching. There are many ways to keep track of discovered websites. Pick one that works for you. I usually start a Microsoft Word document and copy and paste web links and screenshots to be sorted through later. I do it in Word because these findings will be incorporated into my final Word project document.

I searched for "Farmland data". The *first* thing that came up is below. Remember, paid-advertising sites are usually listed first. This one was not an ad. Ads and dot-com sites put me on the defensive: what are they trying to sell me, in this case data? But this is a dot-org site, usually some sort of not-for-profit organization. They obviously have an agenda, saving farmland, which is noble, but would they alter data or selectively withhold or otherwise manipulate data analysis unethically in support of their cause? Reminder to myself, I must do my due diligence if I use data from this source.

https://farmlandinfo.org/data-and-statistics/



Exploring, I found lots of words in the comprehensive and technical reports below, but little data, despite the big "Data and Statistics" in the screenshot above. The report below was helpful in writing my term paper, but I still couldn't find my data.

https://farmlandinfo.org/publications/farms-under-threat-the-state-of-the-states/



The technical report below gave me hope that it would answer all my data due-diligence checklist items. I must remember to refer to this later.

https://farmlandinfo.org/publications/farms-under-threat-technical-report/



Somewhere in my searching these sites, I found a disclaimer about using their data, but forgot to paste the link. All I remember is that it said something about commercial use. Will I be able to find it again?

Eventually I found this link to possible data.

https://farmlandinfo.org/statistics/census-of-agriculture



All producers / primary producers with 10 years or less on any farm				908,274 / 472,360
Market value of agricultural products sold ($1,000)	200,646,355	297,220,491	394,646,980	388,522,695
*Reflects new terms in 2017 Census of Agriculture				

For more information

- FIC 2017 Census of Agriculture Fact Sheet
- Download: Table of Land in Farms by State
- Infographic: All Producers by Age
- Infographic: All New and Beginning Producers by Age
- FIC Highlights: 2017 Census of Agriculture
- Census of Agriculture Website

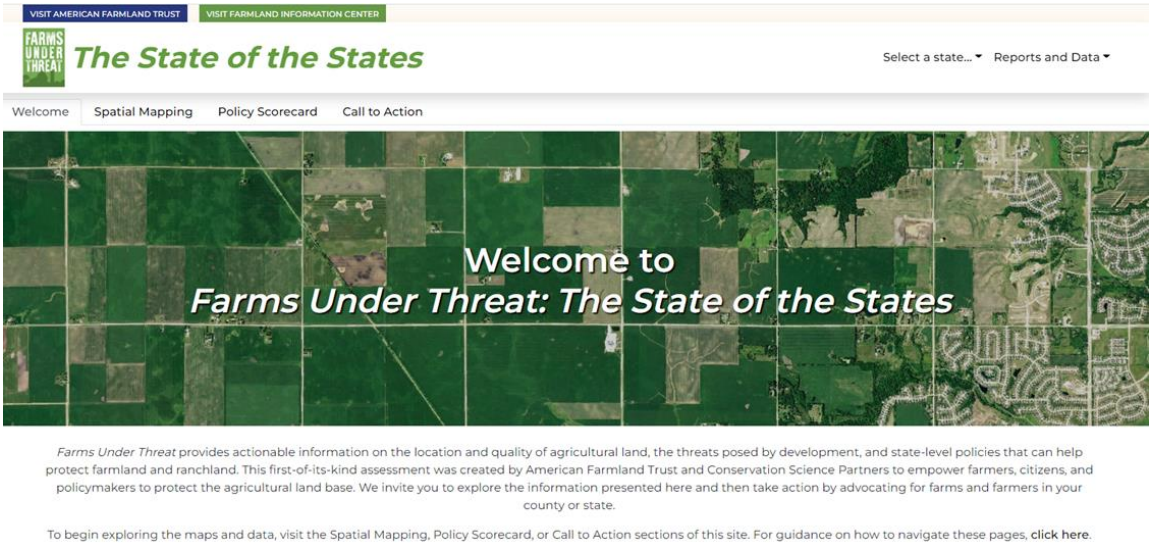
Other Data Sources

- Bureau of Economic Analysis
- Bureau of Labor Statistics
- Foreign Agricultural Trade
- U.S. Census Bureau
- Quick Facts

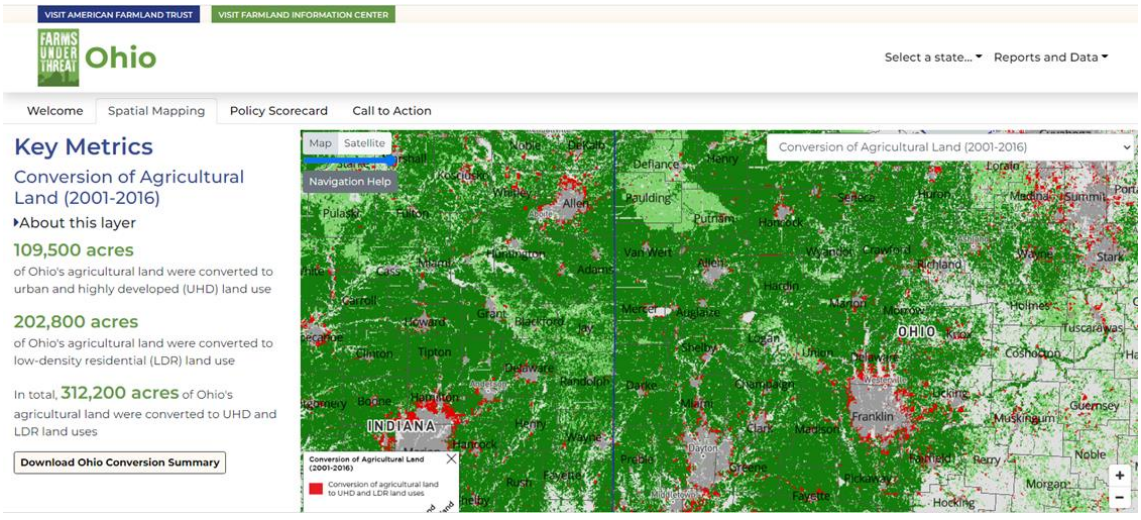
Click on the second link above under "For More Information, "Download Table of Land in Farms by State". Notice the Excel filename.

Finding this link proved to be key.

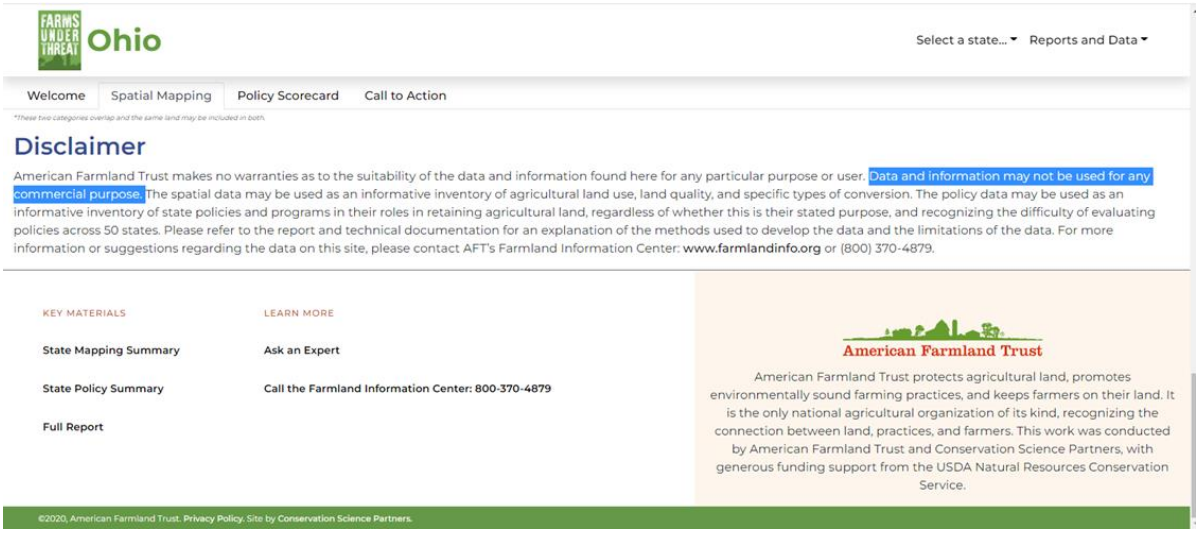
https://csp-fut.appspot.com/



Click "Select a State" (upper right). Then select Ohio.



Scrolling down the page farther yields the screenshot below.



I went back to this screen.

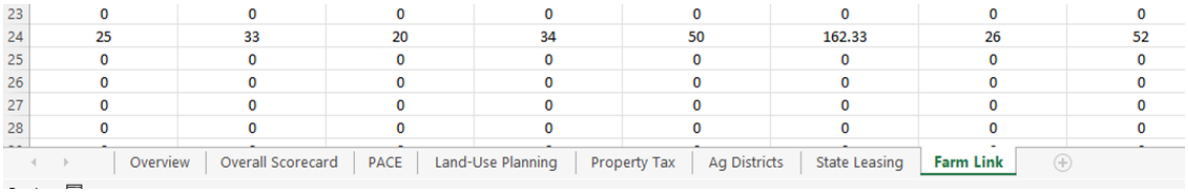
`https://csp-fut.appspot.com/`

Select Reports and Data (upper right). Click the sixth item: "APL Scorecard and Scoresheets (Excel)".

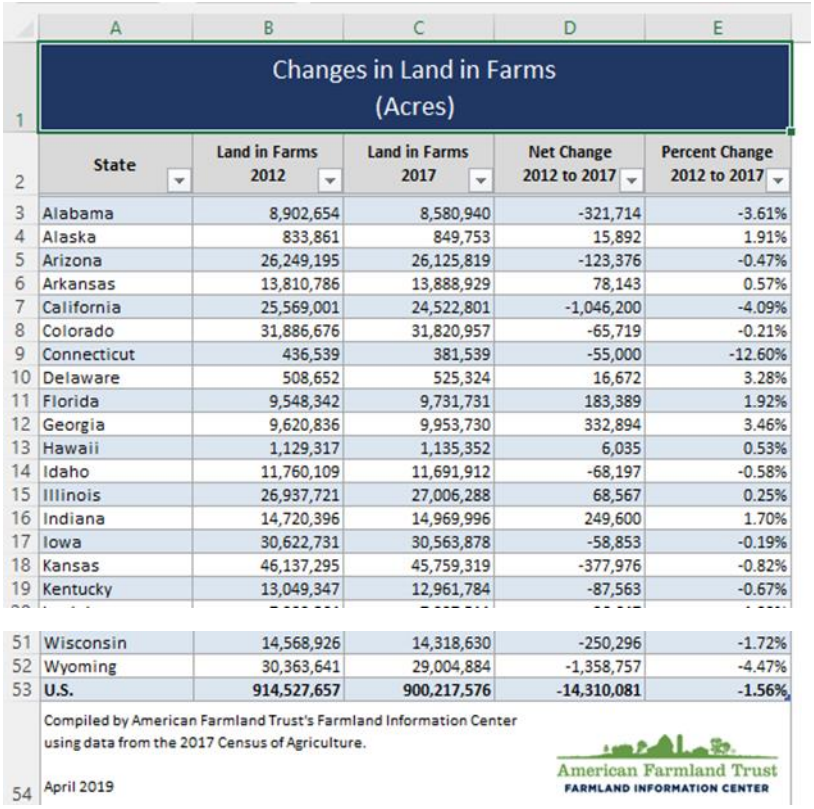
Notice the downloaded Excel file name (lower left corner). `AFT_Ag_Land_Prot…xlsx` (The `.xlsx` indicates Excel file format. This is the format that is often used for downloadable data.)

Overall, I downloaded two Excel files. One with eight sheets and lots of data, none of which was particularly helpful.

Excel File #1: not useful to me.



Excel file #2. This is exactly what I was looking for!



	A	B	C	D	E
1	Changes in Land in Farms (Acres)				
2	State	Land in Farms 2012	Land in Farms 2017	Net Change 2012 to 2017	Percent Change 2012 to 2017
3	Alabama	8,902,654	8,580,940	-321,714	-3.61%
4	Alaska	833,861	849,753	15,892	1.91%
5	Arizona	26,249,195	26,125,819	-123,376	-0.47%
6	Arkansas	13,810,786	13,888,929	78,143	0.57%
7	California	25,569,001	24,522,801	-1,046,200	-4.09%
8	Colorado	31,886,676	31,820,957	-65,719	-0.21%
9	Connecticut	436,539	381,539	-55,000	-12.60%
10	Delaware	508,652	525,324	16,672	3.28%
11	Florida	9,548,342	9,731,731	183,389	1.92%
12	Georgia	9,620,836	9,953,730	332,894	3.46%
13	Hawaii	1,129,317	1,135,352	6,035	0.53%
14	Idaho	11,760,109	11,691,912	-68,197	-0.58%
15	Illinois	26,937,721	27,006,288	68,567	0.25%
16	Indiana	14,720,396	14,969,996	249,600	1.70%
17	Iowa	30,622,731	30,563,878	-58,853	-0.19%
18	Kansas	46,137,295	45,759,319	-377,976	-0.82%
19	Kentucky	13,049,347	12,961,784	-87,563	-0.67%
51	Wisconsin	14,568,926	14,318,630	-250,296	-1.72%
52	Wyoming	30,363,641	29,004,884	-1,358,757	-4.47%
53	**U.S.**	**914,527,657**	**900,217,576**	**-14,310,081**	**-1.56%**
54	Compiled by American Farmland Trust's Farmland Information Center using data from the 2017 Census of Agriculture. April 2019				

I was persistent and found what I want. But next is due diligence on source, licensed usage and accuracy/reliability/bias.

Data Due Diligence

Just because it's on the internet doesn't meant it's true. Just because it's on the internet doesn't mean I can use it. I need to perform my due diligence, including:

- Suitability for my intended purposes
- Completeness
- Accuracy
- Integrity
- Bias
- Currency (I want the latest available)
- Source(s)
- License to use (legality)
- Ethical use of the data

Suitability for my intended purposes:

I think it is exactly what I want for my term paper

Completeness

All 50 states are populated with data in columns B and C. (Columns D and E are calculated from B and C).

Accuracy

Columns B and C are from USDA (see Sources below). Accordingly, they are as accurate as USDA makes them. They use their own rigorous protocols. Surely everybody relies on USDA data. Accordingly, it is safe to assume that the numbers are as accurate as exist. Columns D and E can be checked for accuracy by a few simple calculator calculations to confirm the percentages. (In Excel you create a formula and copy it down columns D and E. Accordingly, in Excel, all formulas will either be consistently right or consistently wrong).

Integrity

How closely does the data I'm using correspond to its original source (USDA NASS database)? Presumably and hopefully it is a 100% match. No differences between original and AFT's use thereof. One assumes that AFT used appropriate data extract, transform, and load (ETL) techniques. But what if AFT used a new hire, lowly paid, computer novice and gave them minimal instructions, without any supervision or audit of the resulting work? Imagine the possibilities for corrupting the data:

- Not knowing any better way, they retyped columns A, B, C from NASS. Imagine the errors.
- What if the NASS data was in state postal code order (AL etc.). They decided to replace these codes with full state names (Alabama, etc). Unfortunately, these lists are not in the same alphabetical order (MA comes before ME, but Maine comes before Massachusetts). Woops. They unknowingly interchanged their data!
- What about DC, Puerto Rico, and other US territories?

Bias

The same comments about columns B and C as above in accuracy, apply here. For columns D and E, if computed correctly, there is no bias in percent changes. The formula is percent change (or percent difference) = (new minus old) / old times 100, for percent form. Ultimately, how can there be bias in "factual" data…the number of acres of farmland? It either is or it isn't…it's not biased. BUT there could be reporting or recording errors. But that comes back to USDA's rigorous data protocols (see above).

Sources:

Whose data is it? Who can authorize or withhold my using the spreadsheet?

If you look at all the URLs above, you can see that there are several domains. In addition, there was farmland.org (the home site of AFT). Whose data is it anyway? The URLs all seem related and also link to each other. Ultimately, I concluded that all the URLs are associated with AFT.

Ultimate Source of the Data: If you read row 54 of the blue spreadsheet above, it says.

"Compiled by American Farmland Trust's Farmland Information Center using data from the 2017 Census of Agriculture. April 2019"

From this, I infer that American Farmland Trust (AFT) added value (analysis) to USDA census data. USDA data (columns B and C) is in the public domain, so I can freely use it in any ethical/legal way. But what about the AFT value added (presumably columns D and E)? Well, anybody is capable of calculating percent change.

Currency

I want the latest data available.

2017 is not very current. But remember from above, USDA conducts its census every five years in years ending in "2" and "7". Accordingly, 2017 and 2012 are the most current numbers available (as of April 2022, the date of this document).

License to use

I struggled with this one because there are several considerations. Columns B and C are in the public domain. Columns D and E are value added by AFT (maybe? Maybe they are from NASS?). But anybody is capable of calculating these percents. In fact, I probably will do my own calculations based on only public domain columns B and C.

But I'm still concerned about the media representation of the data. Is that somehow copyrighted by AFT and thus subject to their license terms? And the spreadsheet was at a different URL than AFT's. But I already inferred that all these sites are affiliated with AFT. And I noticed that AFT's logo is on the blue spreadsheet at the bottom (see above blue Excel screenshot). And it says *"compiled by"*. Are complications of public-domain data separately copyrightable?

And then, finally, I remembered reading something about commercial licensing. Oh yes, please see Ohio screenshot above. This affiliated AFT page says: *"Data and information may not be used for any commercial purpose."* From this, most people would probably infer that non-commercial use is ok. That might be correct, but it does not logically follow. It says nothing about non-commercial use.

Well:

- I'm not using it commercially
- the original data is USDA sourced (public domain)
- AFT's value-added is minimal (and purely mathematical)
- AFT's value-added is not in any way intellectually unique (percent analysis is the most basic form of analysis that everybody knows how to do)
- my usage does not diminish the potential market value of AFT's value-added to the data

Ethical Use of the Data

Above is legal. Legal and ethical are not identical. Two ethical considerations, as follows:

1. Is the data itself ethical?

I have no reason to believe that this data was stolen or misappropriated or created fraudulently prior to my acquiring it (based on the discussions above). Certainly farmland acreage is objective and factual, without hateful or non-DEI attributes.

2. Is my use of the data ethical?

I am going to use the data ethically in support of my term paper which seeks to make recommendations on how to achieve:

- United Nations Sustainable Development Goal #2 Zero Hunger
- Reduce Food insecurity
- Reduce Malnutrition
- Reduce Starvation
- Improve Equity

- Improve Inclusion
- Improve Social justice

I am not going to (mis)use the data in a biased way to mislead readers or to incite hate or non-DEI objectives.

Therefore, I conclude that the data itself and my use thereof are ethical.

Your Project

Part 1

Data project based on USDA Farmland Census Data (columns A, B, C in the above blue Excel screenshot in above spreadsheet).

Using the `Farmland.csv` file, do the following:

- Create a new Jupyter notebook and load the data file into a Pandas DataFrame.
- Based on columns B and C, calculate the absolute difference from 2012 to 2017, and the relative difference (percent change) from 2012 to 2017.
- Sort the data from highest absolute difference to lowest.
- Sort the data from highest relative difference to lowest.
- Calculate total, average, minimum, and maximum for the four data columns.
- Draw four horizontal bar charts, sorted highest to lowest, for the four data columns.

Part 2

While incorporating my data into my term paper, I realized it would be helpful to show readers how important agriculture is to each state. I thought of simply ranking each state's farmland acreage. But using each state's farm acreage as a proxy for agricultural importance won't work because, for example, Texas (a big state) is hundreds of times bigger than Rhode Island (a small state). So of course, Texas's farm acreage will be more.

I next thought of trying to find each state's agricultural component of state GDP (Gross Domestic Product). I decided against it when I considered my audience's likely knowledge base. My term paper is aimed at student environmental activists, not economists.

Ultimately, I decided that each state's farm acreage compared to its total landmass would be an easy-to-understand and (hopefully) easy-to-find proxy for each state's agricultural importance.

BUT, which landmass should I find? 2012 or 2017? On reflection, silly me. Each state's landmass is unchanged from when it was first admitted, Hawaii becoming the 50th state on August 21, 1959. Accordingly, all listings of landmass-by-state subsequent to that date should be the same.

Your Data Collection Task

Find a list of each state's total landmass. Hopefully your source(s) will have some authoritativeness. Hopefully it will be in the same alphabetical order as in the data from Part 1 above. Hopefully it will be in acres and not square miles. Hopefully you can manipulate it. Hopefully you don't end up copying and pasting 50 numbers (or worse, retyping them).

If you find square miles (likely), include in your Python coding a conversion formula to acres.

Your Data Merging Task

Having found your state landmass data (converted from square miles to acres, if necessary), incorporate it into your analysis, to produce these additional results:

- Calculate each state's 2012 and 2017 farmland as a percent of total state landmass
- Calculate a percent of total US landmass
- Sort the data from highest to lowest percent of total state landmass

Your Data Analysis Task

Using Markdown syntax, write a few paragraphs in your Jupyter notebook on what you have learned about both (1) data issues and (2) what the data is telling you about the status of farmland in USA and policy questions that might be raised.